Homework 1

Computer Simulation 1

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Problem 1: Statement

Calculate the exact solution of the one-dimensional Ising model with N spins for free boundary conditions:

a) Sketch the calculation of the partition function Z by successive summation of spins.

b) Calculate the internal energy $U = \langle E \rangle = -\dfrac{\partial \ln Z}{\partial \beta}$ and the specific heat $C = \dfrac{dU}{dT}$.

c) How do $u = U/N$ and $c = C/N$ behave in the thermodynamic limit $N \to \infty$?

Problem 1: Solution

Part (a)

Let us consider a one dimensional Ising model with N spins for free boundary conditions. We can write the energy of such system as follows

$$E = -J \sum_{i=1}^{N-1} s_i s_{i+1} \tag{1}$$

where the sum is over all pairs of neighbouring spins i and $i+1$ and J is an exchange constant assumed to be positive. For algebraic simplicity's sake, we can expand eq. (1) as follows

$$E = -J(s_1 s_2 + s_2 s_3 + \cdots + s_{N-2} s_{N-1} + s_{N-1} s_N) \tag{2}$$

Furthermore, by definition, we know that the partition function is given by the following equation:

$$Z = \sum_{\{s_i = \pm 1\}} e^{\frac{-E}{k_B T}} \tag{3}$$

Plugging in eq. (2) into eq. (3) we arrive at eq. (4)

$$Z = \sum_{\{s_i=\pm 1\}} e^{\frac{J}{k_B T}(s_1 s_2 + s_2 s_3 + \cdots + s_{N-2} s_{N-1} + s_{N-1} s_N)} \tag{4}$$

This can be expanded over all states as follows:

$$Z = \sum_{s_1=\pm 1} \cdots \sum_{s_N=\pm 1} e^{\frac{J}{k_B T}(s_1 s_2 + s_2 s_3 + \cdots + s_{N-2} s_{N-1} + s_{N-1} s_N)} \tag{5}$$

Notice how the term s_N appears only in the last term. Therefore, we can extract the sum accounting for the last term as shown in eq. (6)

$$Z = \sum_{s_1=\pm 1} \cdots \sum_{s_{N-1}=\pm 1} e^{\frac{J}{k_B T}(s_1 s_2 + s_2 s_3 + \cdots + s_{N-2} s_{N-1})} \sum_{s_N=\pm 1} e^{\frac{J}{k_B T} s_{N-1} s_N} \tag{6}$$

Since s_N can only be $+1$ or -1, we calculate the last term as:

$$\sum_{s_N=\pm 1} e^{\frac{J}{k_B T} s_{N-1} s_N} = e^{\frac{J}{k_B T} s_{N-1}} + e^{-\frac{J}{k_B T} s_{N-1}} \tag{7}$$

Recalling the hyperbolic cosine definition given by eq. (8)

$$\cosh(\theta) = \frac{e^{\theta} + e^{-\theta}}{2} \tag{8}$$

One can rewrite eq. (7) in terms of the hyperbolic cosine. Since we know that $s_{N-1} = \pm 1$ and since the hyperbolic cosine is an even function we can further simplify the expression as follows

$$\sum_{s_N=\pm 1} e^{\frac{J}{k_B T} s_{N-1} s_N} = 2\cosh\left(\frac{J s_{N-1}}{k_B T}\right) = 2\cosh\left(\frac{J}{k_B T}\right) \tag{9}$$

Plugging this back into eq. (6)

$$Z = \sum_{s_1=\pm 1} \cdots \sum_{s_{N-1}=\pm 1} e^{\frac{J}{k_B T}(s_1 s_2 + s_2 s_3 + \cdots + s_{N-2} s_{N-1})} 2\cosh\left(\frac{J}{k_B T}\right) \tag{10}$$

we realize this can be done recursively from the "back" of the expression towards the front, i.e.:

$$Z = \sum_{s_1=\pm 1} \cdots \sum_{s_{N-2}=\pm 1} e^{\frac{J}{k_B T}(s_1 s_2 + s_2 s_3 + \cdots + s_{N-3} s_{N-2})} 2\cosh\left(\frac{J}{k_B T}\right) 2\cosh\left(\frac{J}{k_B T}\right) \tag{11}$$

$$Z = \sum_{s_1=\pm 1} \cdots \sum_{s_{N-3}=\pm 1} e^{\frac{J}{k_B T}(s_1 s_2 + s_2 s_3 + \cdots + s_{N-4} s_{N-3})} 2\cosh\left(\frac{J}{k_B T}\right) 2\cosh\left(\frac{J}{k_B T}\right) 2\cosh\left(\frac{J}{k_B T}\right) \tag{12}$$

and so on, $N-1$ times, until you arrive at the following

$$Z = \left[2\cosh\left(\frac{J}{k_B T}\right)\right]^{N-1} \sum_{s_1=\pm 1} 1 \tag{13}$$

and since $\sum_{s_1=\pm 1} 1 = 1 + 1 = 2$, finally we get:

$$\boxed{Z = 2^N \left[\cosh\left(\frac{J}{k_B T}\right)\right]^{N-1}} \tag{14}$$

Part (b)

Now we are tasked with calculating the internal energy $U = \langle E \rangle = -\frac{\partial \ln Z}{\partial \beta}$ and the specific heat $C = \frac{dU}{dT}$.

First we calculate the internal energy by taking a partial derivative with respect to β, which is assumed to be equal to $\beta = \frac{1}{k_B T}$.

$$\begin{aligned} U &= -\frac{\partial \ln Z}{\partial \beta} = -\frac{\partial}{\partial \beta} \ln\left(2^N \left[\cosh(J\beta)\right]^{N-1}\right) \\ &= -\frac{\partial}{\partial \beta}\left(N \ln 2 + (N-1)\ln\left(\cosh(J\beta)\right)\right) \\ &= -\frac{\partial}{\partial \beta}(N-1)\ln\left(\cosh(J\beta)\right) \end{aligned}$$

by employing the chain rule we get

$$= -(N-1)\frac{1}{\cosh(J\beta)}\sinh(J\beta)J$$

furthermore using the definition of the hyperbolic tangent we conclude

$$\boxed{U = -J(N-1)\tanh(J\beta)} \tag{15}$$

or equivalently

$$\boxed{U = -J(N-1)\tanh\left(\frac{J}{k_B T}\right)} \tag{16}$$

If we wish to calculate the specific heat, we take a derivative of the internal energy function with respect to temperature.

$$\begin{aligned} C = \frac{dU}{dT} &= \frac{d}{dT}\left(-J(N-1)\tanh\left(\frac{J}{k_B T}\right)\right) \\ &= -J(N-1)\frac{d}{dT}\tanh\left(\frac{J}{k_B T}\right) \end{aligned}$$

by knowing that $\frac{d\tanh(\theta)}{d\theta} = \operatorname{sech}^2(\theta)$ we conclude

$$\boxed{C = \frac{J^2}{k_B T^2}(N-1)\operatorname{sech}^2\left(\frac{J}{k_B T}\right)} \tag{17}$$

Part (c)

Finally, we observe how $u = U/N$ and $c = C/N$ behave in the thermodynamic limit $N \to \infty$

$$u = \frac{-J(N-1)\tanh\left(\frac{J}{k_B T}\right)}{N}$$

$$c = \frac{\frac{J^2}{k_B T^2}(N-1)\operatorname{sech}^2\left(\frac{J}{k_B T}\right)}{N}$$

taking the limit as $N \to \infty$ for both expressions

$$\begin{aligned}
&\lim_{N\to\infty} \frac{-J(N-1)\tanh\left(\frac{J}{k_B T}\right)}{N} \\
&= -J\tanh\left(\frac{J}{k_B T}\right)\lim_{N\to\infty}\frac{N-1}{N} \\
&= -J\tanh\left(\frac{J}{k_B T}\right)\lim_{N\to\infty}\left(1-\frac{1}{N}\right) \\
&\lim_{N\to\infty} u = \boxed{-J\tanh\left(\frac{J}{k_B T}\right)}
\end{aligned} \tag{18}$$

$$\begin{aligned}
&\lim_{N\to\infty} \frac{\frac{J^2}{k_B T^2}(N-1)\operatorname{sech}^2\left(\frac{J}{k_B T}\right)}{N} \\
&= \lim_{N\to\infty} \frac{J^2}{k_B T^2}\operatorname{sech}^2\left(\frac{J}{k_B T}\right)\frac{N-1}{N} \\
&= \frac{J^2}{k_B T^2}\operatorname{sech}^2\left(\frac{J}{k_B T}\right)\lim_{N\to\infty}\frac{N-1}{N} \\
&= \frac{J^2}{k_B T^2}\operatorname{sech}^2\left(\frac{J}{k_B T}\right)\lim_{N\to\infty}\left(1-\frac{1}{N}\right) \\
&\boxed{\lim_{N\to\infty} c = \frac{J^2}{k_B T^2}\operatorname{sech}^2\left(\frac{J}{k_B T}\right)}
\end{aligned} \tag{19}$$

Problem 2: Statement

a) Plot u and c of exercise 1 as function of temperature T (e.g., with "gnuplot") for various finite N $(= 20, 40, \ldots, 100)$ and discuss the curves (for example: Does a maximum exist for c and, if yes, at which temperature?).

b) Study how the thermodynamic limit $N \to \infty$ is approached?

Problem 2: Solution

Part (a)

We simplify expressions given by eq. (16) and (17)

$$u(N,T) = -J\frac{(N-1)}{N}\tanh\left(\frac{J}{k_B T}\right) \tag{20}$$

$$c(N,T) = \frac{J^2}{k_B T^2}\frac{(N-1)}{N}\operatorname{sech}^2\left(\frac{J}{k_B T}\right) \tag{21}$$

Then we plot u as a function of T for several values of N, as demanded in the problem statement.

[16]:
```
import numpy as np
from matplotlib import pyplot as plt

J = 1.0
k_b = 1.0

N_values = [20, 40, 60, 80, 100]

T = np.linspace(0.01, 50, 500)

def u(N,T):
  return -J * (N-1) / N * np.tanh(J/(k_b*T))

for N in N_values:
    plt.plot(T,u(N,T), label=f"N={N}")

plt.xlabel("T")
plt.ylabel("u(N,T)")
plt.legend()
plt.show()
```

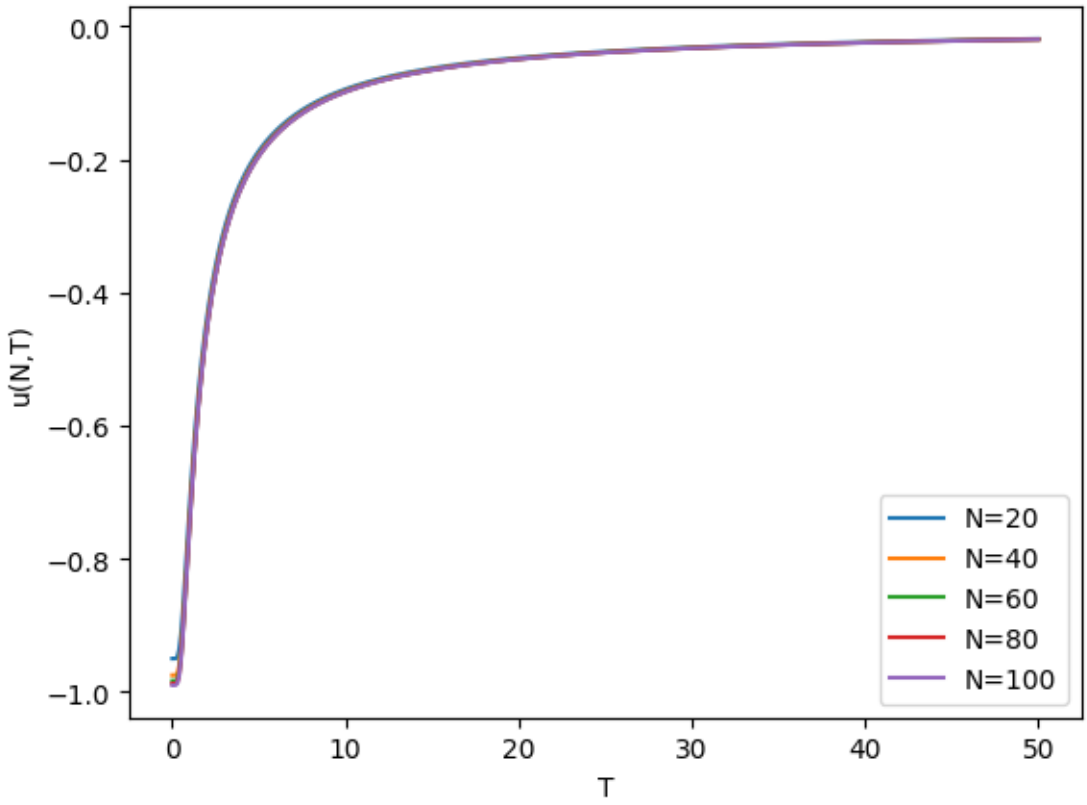


Figure 1. Dependence of $u(N,T)$ on T, for $N = 20, 40, 60, 80, 100$

As we can see on Fig. 1, as $T \to \infty$, $u(N,T) \to 0$. Additionally we see that as N is increased, the curves start off at lower and lower $u(N,T)$. More mathematically speaking as $T \to 0$, $u(N,T) \to -J(N-1)/N$.

Furthermore we plot c as a function of T for several values of N, as stated in the problem statement.

[17]:
```
def c(N, T):
    return J**2/(k_b*T**2) * (N-1)/N * 1/(np.cosh(J/(k_b*T))**2)

for N in N_values:
    plt.plot(T,c(N,T), label=f"N={N}")

plt.xlabel("T")
plt.ylabel("c(N,T)")
plt.legend()
plt.show()
```

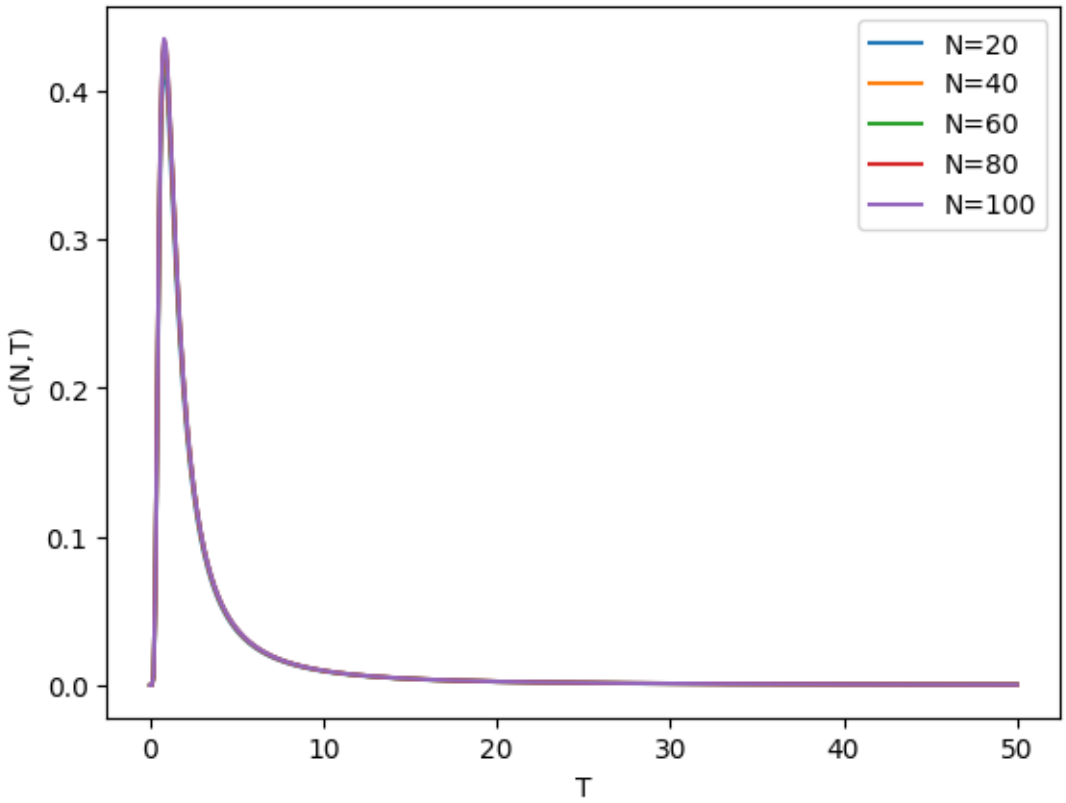


Figure 2. Dependence of $c(N,T)$ on T, for $N = 20, 40, 60, 80, 100$

As we can see on Fig. 2, as $T \to \infty$, $c(N,T) \to 0$. Additionally, there exists a clear maximum of $c(N,T)$. $c(N,T)$ increases from zero, reaches a maximum near $T = 1$, and then decreases towards zero.

Let us plot a plot with a smaller T bandwidth to identify at which exact T value the maximum of $c(N,T)$ appears . We use `np.argmax` to find the largest values of c for each N, and then find the corresponding T at which it appears.

[23]:
```
T_2 = np.linspace(0.01, 5, 500)

for N in N_values:
    c_val = c(N, T_2)
    plt.plot(T_2, c_val, label=f"N={N}")

    i_max = np.argmax(c_val)
    T_max = T_2[i_max]

    print(f"N={N}: T_max={T_max:.4f}")

plt.axvline(0.83, linestyle="--", color="red", label="T=0.83")
```

```
plt.xlabel("T")
plt.ylabel("c(N,T)")
plt.legend()
plt.show()
```

```
N=20: T_max=0.8300
N=40: T_max=0.8300
N=60: T_max=0.8300
N=80: T_max=0.8300
N=100: T_max=0.8300
```

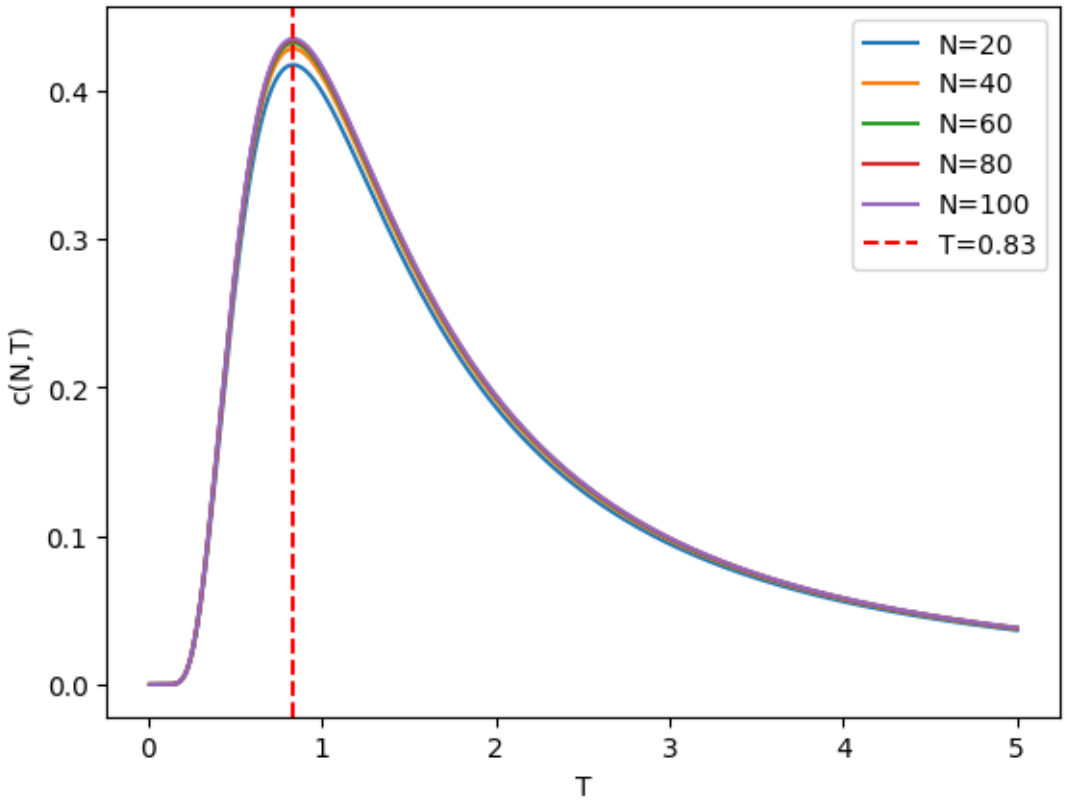


Figure 3. Dependence of $c(N,T)$ on T, for $N = 20, 40, 60, 80, 100$ and $T \leq 5$

On Fig. 3 we can much more clearly see that the maximum of $c(N,T)$ exists at T around 1. Numerically we determine this number to be $T = 0.83$. We also notice that increasing N changes the height of the curves but not the temperature at which the heat capacity reaches its maximum.

Part (b)

Now we wish to study how the thermodynamic limit of $N \to \infty$ behaves for both $u(N,T)$ and $c(N,T)$.

Let us remember eq. (18) and (19) derived in the previous task.

$$\lim_{N\to\infty} u = -J \tanh\left(\frac{J}{k_B T}\right)$$

$$\lim_{N\to\infty} c = \frac{J^2}{k_B T^2} \operatorname{sech}^2\left(\frac{J}{k_B T}\right)$$

It is useful to notice that u and c themselves given by eq. (20) and (21) are nothing more than

$$u(N,T) = \frac{N-1}{N} \lim_{N\to\infty} u$$

$$c(N,T) = \frac{N-1}{N} \lim_{N\to\infty} c$$

Since the limits in eq. (18) and (19) are independent of N, now we see that the thermodynamic limit is approached through the factor $\frac{N-1}{N} = 1 - \frac{1}{N}$. This means that the error from the thermodynamic limit decreases proportionally with $\frac{1}{N}$.